DATA ANALYSIS REPORT

CUSTOMER 360

Prepared by:
Phuc Le

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INTRODUCTION

RFM, also known as **RFM analysis**, is a type of customer segmentation and behavioral targeting used to help businesses rank and segment customers based on the recency, frequency, and monetary value of a transaction

In an era of rapidly advancing technology, tracking and collecting information from various sources and analyzing the potential aspects of customers will help businesses easily reach their target audience. This allows businesses to make well-informed decisions and devise the most effective development strategies.

Therefore, to accurately determine customer needs, businesses must analyze the available data to make the most informed decisions. This is where Customer 360 analysis comes into play. But what exactly is Customer 360 analysis? Customer 360 is simply understood as a system that analyzes all customer information through various aspects. This system allows businesses to have the most detailed view of all the touchpoints where customers interact with them, such as purchase frequency, lifetime value, and more.

If businesses are always present at these touchpoints on the customer journey and quickly collect information to continuously update the Customer 360 portrait, understanding customers and meeting their needs and desires will no longer be a difficult task.

In this analysis, we will focus on Customer Clustering with the RFM Model. Specifically, the RFM Model is one of the models used to analyze customers based on three metrics, with scores evaluated on a scale from 1 to 4, where 1 is the lowest and 4 is the highest.

- **R (Recency)**: The last time the customer used the service
- **F (Frequency)**: The frequency with which the customer uses the service
- **M (Monetary)**: The revenue generated from the service

CHAPTER I

OVERVIEW OF RFM CUSTOMER 360

01 Introducting Customer 360

Customer 360 provides a comprehensive view of customer data by aggregating all information about a customer. It includes not only demographic information but also their relationships, activities, and lifecycle stages with the business.

With the aggregated data, businesses can build complete customer profiles. This helps them gain a holistic view of how their operations impact the customer experience.

All data is stored and shared with company members on a single platform. This allows businesses to create new strategies to attract customers and provide them with the best possible service experiences.

02 Benefit Of Customer 360 Analysis

Customer 360 helps businesses understand their customers, enabling them to provide superior customer experiences and ensure their own competitive advantage. With Customer 360, businesses can:

01 **Gain a comprehensive view of customers**
The shared Customer 360 view allows departments within the business to easily access customer information. This facilitates the creation of perfect business strategies that ensure trustworthy relationships with customers.

02 **Understand customers and their behaviors**
With Customer 360, businesses will have basic information, behaviors, and shopping habits of customers. This enhances customer retention and loyalty by connecting data about their activities and making more informed decisions to meet customer needs. At the same time, it allows businesses to quickly predict new opportunities.

03 **Enhance the brand through customer data**
From a unified view of customers, including purchase history, preferences, and even dissatisfaction, businesses can adjust to better suit similar customer segments. This helps enhance the brand image in the eyes of customers.

04 **Create better sales and marketing programs**
Based on customer information, sales and marketing departments can develop promotional programs that improve customer retention and loyalty.

05 **Streamline business operations**
Automate workflows so that data can be accessed from the start to support customer onboarding, self-scheduling, and other activities related to customers and the business.

03 Fundamental Concepts In RFM Customer 360 Analysis

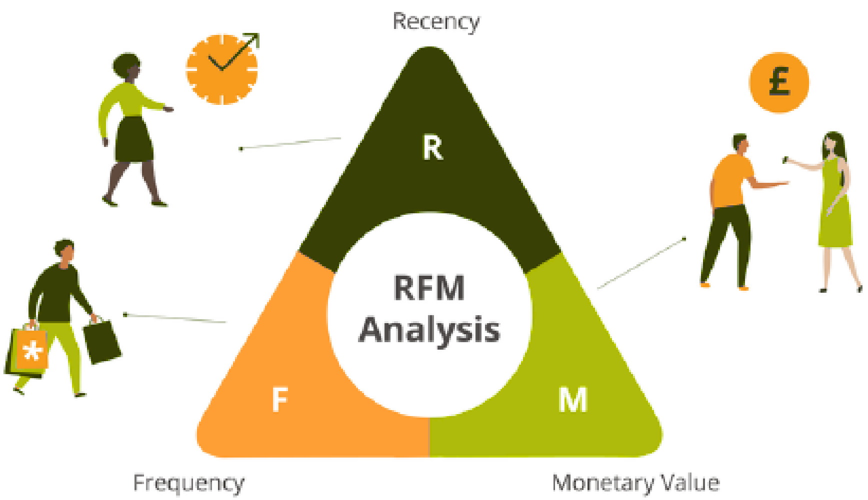
RFM Customer 360 is a customer analysis method based on the RFM (Recency - Frequency - Money) model.

This method helps businesses evaluate the value of each customer based on their level of interaction with the business. From there, businesses can set strategic business efficiency and prioritize maximum profits. Specifically, the three main factors in RFM analysis are as follows:

Recency
The time since the customer's last purchase.

Frequency
The number of times the customer has made purchases within a certain period.

Monetary
The total amount of money spent by the customer



04 Benefit OF RFM Customer 360 Analysis

When implementing RFM analysis, businesses can derive numerous benefits. Specifically:

Enhanced customer segmentation

This approach allows businesses to access more customer data, enabling segmentation and adjustment of marketing programs based on each group's behavior and preferences.

Better understanding of customers

Through RFM, businesses gain insights into recent trends among customers, understanding their preferences, habits, and allowing for tailored marketing strategies for specific segments.

Identification of loyal & high-risk customers

Firstly, identifying loyal customers allows businesses to allocate time and resources more effectively. Secondly, recognizing high-risk customers enables businesses to devise strategies to retain them rather than losing them.

Improved customer retention

By accurately identifying target customers, businesses can offer tailored incentives based on their habits and preferences, thereby increasing customer spending and fostering loyalty.

Enhanced profitability

Utilizing RFM analysis enables businesses to maximize resources towards appropriate customer segments, leading to significant improvements in revenue and profitability.

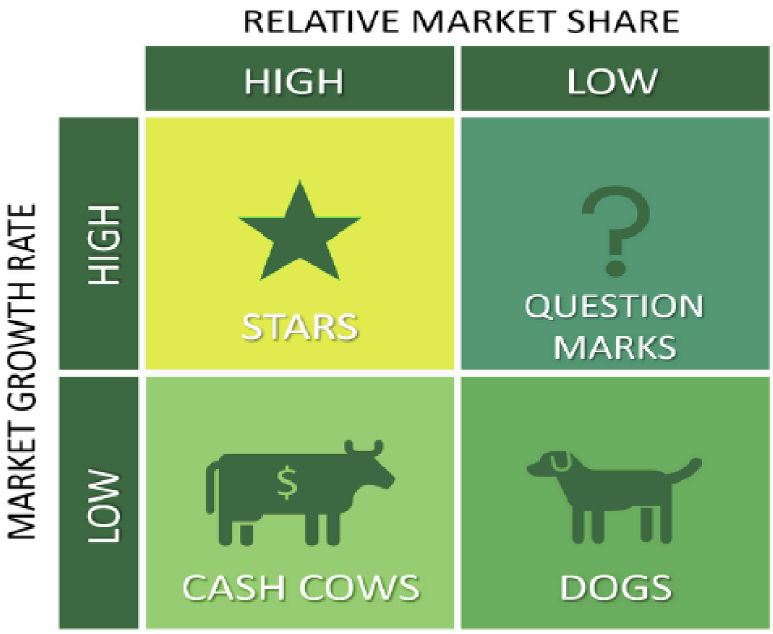
Time and cost savings

RFM analysis allows businesses to identify and focus on customer segments that warrant the investment of time and resources efficiently.

These benefits underscore the strategic importance of RFM analysis in **enhancing customer relationships and optimizing business outcomes**

05 Customer Definition by BCG Matrix

The BCG matrix (also known as the Boston matrix) is a strategic business analysis toll used to evaluate the growth rate and market share of products or product lines. Based on this evaluation, businesses can make decisions to retain, divest, invest more, or eliminate products



The BCC matrix categorizes products into four SBU (Strategic Business Unit) statuses:

- **Star (VIP Customers):** This group of customers generates the main profit for the business, with frequent usage and recent service use close to the current time.
- **Cash Cow (General Customers):** This group has relatively low spending and purchase frequency, but they use the service regularly and have made recent purchases.
- **Question Mark (Potential Customers):** This group consists of customers who make frequent purchases and spend a lot, and their last use of the service was recent, but their usage frequency is not high.
- **Dog (Secondary Customers):** This group has low spending and brings the least profit to the business, with infrequent service usage and the last use being relatively long ago.

CHAPTER II

DETERMINE CUSTOMER GROUPS

01 Data seperation

Description of data sheets

In this project, we have two tables that need to be analyzed:

Customer_Transaction table

Column Name	#	Data Type	Not Null	Auto Increment	Key	Default	Extra
123 Transaction_ID	1	bigint	[ ]	[ ]			
123 CustomerID	2	bigint	[ ]	[ ]			
A-Z Purchase_Date	3	varchar(50)	[ ]	[ ]			
123 GMV	4	bigint	[ ]	[ ]			

Customer_Transaction Table: Shows customer information from June 2022 to August 2022. Each row on the table is each transaction task. The Customer_Transaction table has 4 columns of data. Each transaction will be identified by [ID]. Each transaction records information: Which Customer [CustomerID], purchase date [Purchase_Date], how much money [GMV].

Customer_Registered table

Column Name	#	Data Type	Not Null	Auto Increment	Key	Default	Extra
123 ID	1	bigint	[v]	[ ]			
A-Z Contract	2	varchar(50)	[ ]	[ ]			
123 LocationID	3	int	[ ]	[ ]			
123 BranchCode	4	tinyint	[ ]	[ ]			
123 Status	5	tinyint	[ ]	[ ]			
A-Z created_date	6	varchar(50)	[ ]	[ ]			
A-Z stopdate	7	varchar(50)	[ ]	[ ]			

Customer_Registered Table: Display the profiles of customers who have registered for membership cards. In this case, we will use the information from the Customer_Registered table to calculate the contract age of the customer, so that we could normalize FREQUENCY and MONETARY values by time (contract age) so customers with different lifespans can be fairly compared

02 Build RFM SCORE scoring framework and classification in SQL

Tool: MySQL

How to do:

- Process OTLP data to OLAP from Customer_Transaction table
- Join two tables Customer_Transaction and Customer_Registered to get information
- To process we will use SQL techniques: WINDOW FUNCTIONS, CTEs, CASE WHEN, DATEDIFF

Step 1: Calculation: Recency, Frequency, Monetary

Recency
Length of time since the customer's last purchase

Frequency
Total number of purchases made by a customer over their lifetime

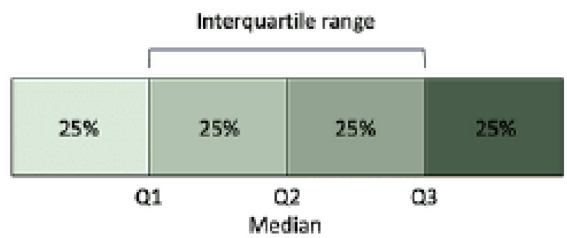
Monetary
Total revenue from customers

CustomerID	Recency	Frequency	Monetary
76,399	92	0.1429	10,000
76,551	92	0.1429	15,000
76,583	92	0.1429	15,000
76,782	92	0.1429	15,000
77,089	92	0.1429	15,000
77,821	92	0.1429	15,000
78,008	92	0.1429	14,285.7143
78,209	92	0.1429	13,571.4286
78,394	92	0.1429	13,571.4286
78,470	92	0.1429	15,000
78,543	92	0.1429	15,000
78,644	92	0.1429	13,571.4286
78,889	92	0.1429	15,000
79,050	92	0.1429	13,571.4286
79,204	92	0.1429	13,571.4286
79,364	92	0.1429	4,285.7143
79,432	92	0.1429	13,571.4286
79,503	92	0.1429	5,000

Step 2: Score RFM according to IQR in SQL

Divide the RFM metrics into four equal groups (quartiles), assigning scores from 1 to 4. A higher score indicates better customer value

Note: For Recency, the scoring is reversed: customers with fewer days since their last purchase receive higher scores.



CustomerID	Recency	Frequency	Monetary	R	F	M
76,399	92	0.1429	10,000	1	1	1
76,551	92	0.1429	15,000	1	1	1
76,583	92	0.1429	15,000	1	1	1
76,782	92	0.1429	15,000	1	1	1
77,089	92	0.1429	15,000	1	1	1
77,821	92	0.1429	15,000	1	1	1
78,008	92	0.1429	14,285.7143	1	1	1
78,209	92	0.1429	13,571.4286	1	1	1
78,394	92	0.1429	13,571.4286	1	1	1
78,470	92	0.1429	15,000	1	1	1
78,543	92	0.1429	15,000	1	1	1
78,644	92	0.1429	13,571.4286	1	1	1
78,889	92	0.1429	15,000	1	1	1
79,050	92	0.1429	13,571.4286	1	1	1
79,204	92	0.1429	13,571.4286	1	1	1
79,364	92	0.1429	4,285.7143	1	1	1
79,432	92	0.1429	13,571.4286	1	1	1
79,503	92	0.1429	5,000	1	1	1
79,575	92	0.1429	12,142.8571	1	1	1
79,695	92	0.1429	12,142.8571	1	1	1
79,711	92	0.1429	15,000	1	1	1
79,794	92	0.1429	15,000	1	1	1
79,803	92	0.1429	15,000	1	1	1
79,931	92	0.1429	7,142.8571	1	1	1
79,937	92	0.1429	7,142.8571	1	1	1

Step 2: Score RFM according to IQR in SQL

We will categorize customers using the standard classification method into seven different groups, including:

- Superiors Loyalists
- Potential Loyalists
- Responsive
- Promising
- Hibernating Loyalists
- Need Attention
- About to Sleep
- At Risk

	RFM_Segment
1	About to Sleep
2	At Risk
3	Hibernating Loyalists
4	Need Attention
5	Potential Loyalists
6	Promising
7	Responsive
8	Superriors Loyalists

Step 2: Score RFM according to IQR in SQL

Customer segments	RFM score group	Characteristics
Superiors Loyalists	344, 443, 444, 343	They are highly loyal customers who engage positively, have high purchase frequency, and generate significant revenue for the business.
Potential Loyalists	442, 441, 342, 341	Customers with relatively high purchase frequency and stable spending on the company's products or services.
Responsive	313, 423, 424, 324, 414, 323, 314	Customers who, despite lower purchase frequency, generate significant revenue. They are highly responsive to offers and promotions.
Promising	412, 411, 421, 312, 321, 311, 422, 322	Customers with the most recent purchases. However, their frequency and revenue are not yet high, but they have strong growth potential.
Hibernating Loyalists	244, 144, 143, 243	Previously active customers who have slowed down in engagement. Targeted re-engagement efforts (e.g., discounts or "We miss you" campaigns) may help.
Need Attention	223, 123, 114, 213, 224, 113, 214, 124	Customers with low engagement and moderate revenue. They require attention to prevent further decline.
About to Sleep	241, 141, 242, 142	Previously frequent buyers who are now declining in activity and revenue. Retention strategies are needed.
At Risk	122, 212, 121, 211, 221, 222, 111, 112	Customers with low engagement, low purchase frequency, and low revenue. Immediate action is needed to understand and improve their behavior.

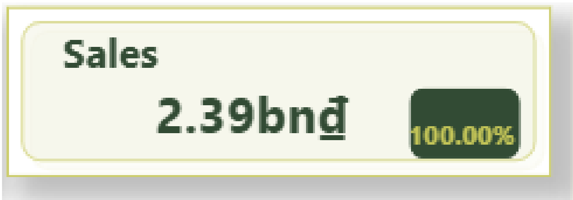
CHAPTER III

CUSTOMER ANALYSIS

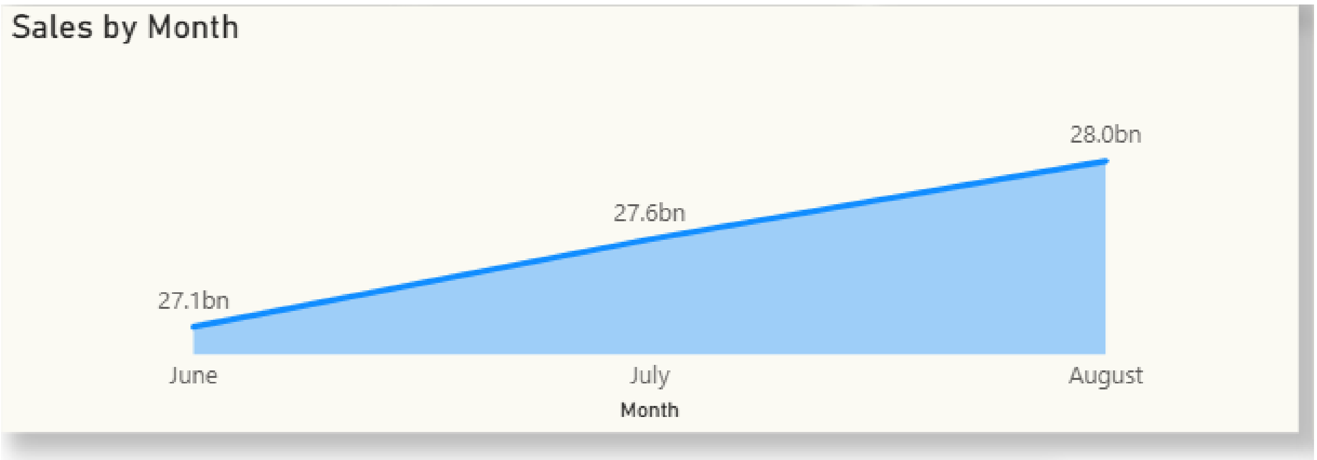
01 Overview of customer

Using **Power BI**

Currently, the database records show that more than 114K0 customers are using the service. Additionally, the revenue generated by these customers has readched 2.39 billion VND



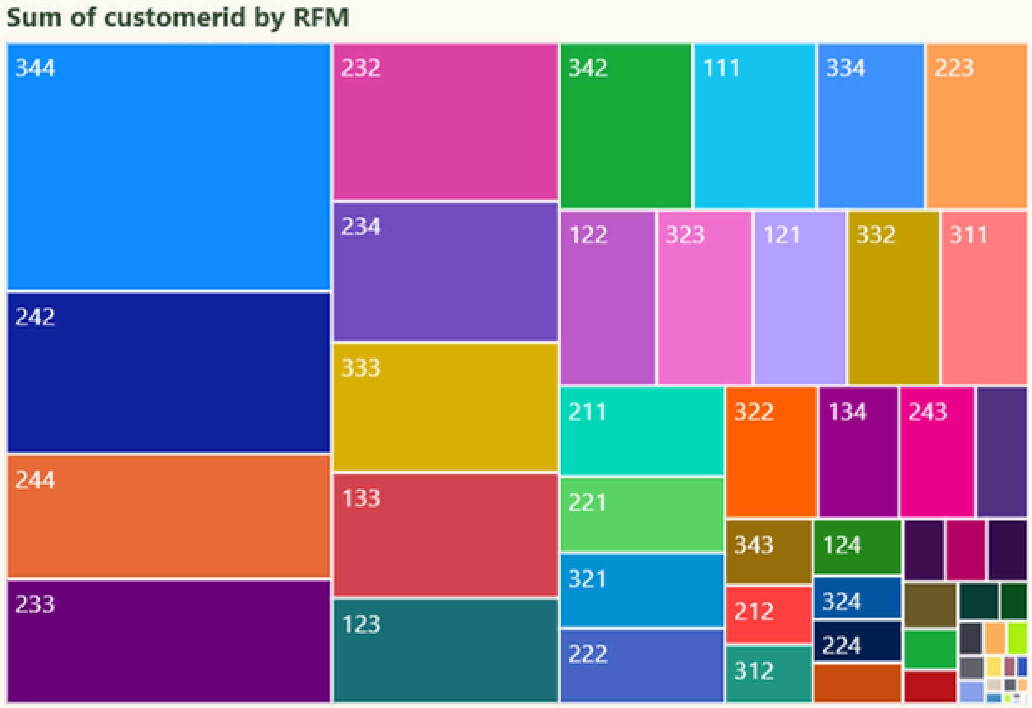
Additionally, the growth rate has shown significant improvement month over month. However, to maximize profitability for the business, we need to identify and analyze customer segments to understand which ones are the most impacting.



CHAPTER III

Next, let's take an overvire of the customer situation by different groups according to the chart below:

RFM	No_Customer
344	12696
111	7558
242	5761
311	5655
211	5513
233	5307
244	4650
232	4381
123	4261
234	3991
333	3948
133	3922
121	3336
122	3066
223	3060
323	3026
[illegible]	[illegible]
Total	**114081**



Looking at the overview chart of customer statistics, we can see the distribution among different customer groups. Group 344 has the highest number of customers. Following that are groups 111,242,311, 211 and 233 with relatively large number of customers ranging from 5,000 to 7,500.

On the other hand, there are groups with very few customers, such as 142, 413, 432, 433,422 and 412 each having only 5,4 and 2 customers respectively.

From this, we can see that customers with a high percentage are usually those with a very **high purchase frequency** and **high value of purchase**. The revenue the company can obtain from this customer group is relatively high.

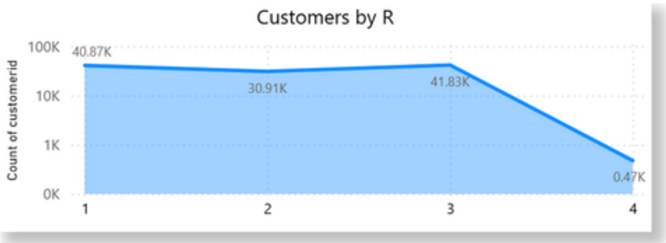
Conversely, a low percentage of customer groups with **a recency is high but low in frequency and monetary**, showing potentially room to improve

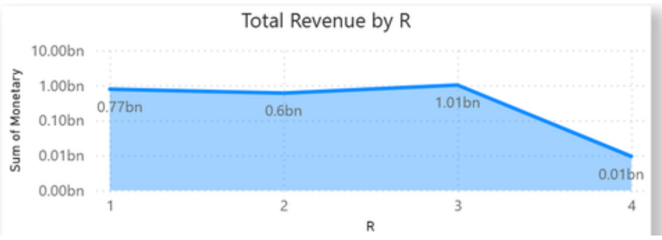
02 Analyze customers from different aspects

To evaluate the customer groups in more detail, let's analyze each aspect of R, F, and M as follows:

From the RFM Model, we have over 60 different customer groups analyzed based on three aspects:

- How frequently do customers use the service?
- When was the last time customers used the service or made a purchase?
- How much money do customers spend on the company's products?





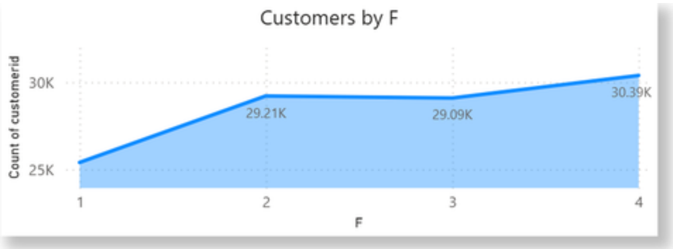
Recency reflects how recently customers made their last purchase, with lower scores indicating longer inactivity. From the chart, group 4 (most recent customers) has a significantly smaller number of customers compared to the other groups. This suggests that the business is currently struggling to attract and retain active customers, as fewer customers are making recent purchases.

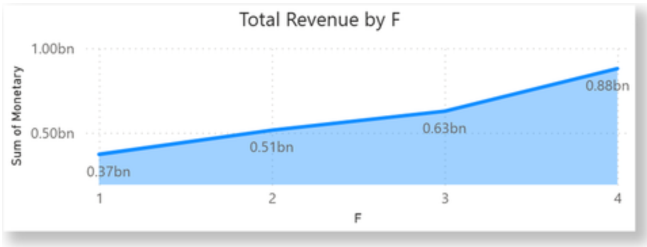
Revenue is concentrated in less recent customer groups (1 and 2), with a ratio of 1.37 to 1.02 compared to more active groups. This suggests the business is relying on customers who are not actively engaging, posing a risk of revenue decline if they churn.

The business should **prioritize re-engagement and retention strategies** to bring customers back. This includes launching personalized campaigns for inactive users, improving product offerings to increase relevance, and implementing loyalty programs to encourage repeat purchases. Additionally, using **targeted promotions and remarketing based on past behavior** can help reactivate customers and strengthen long-term engagement.

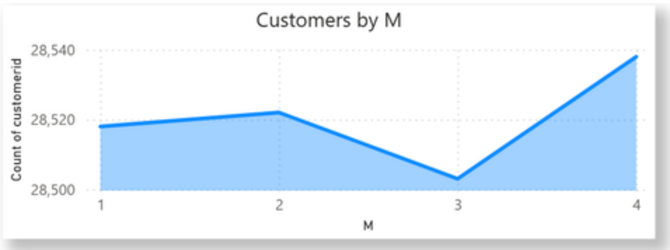
02 Analyze customers from different aspects

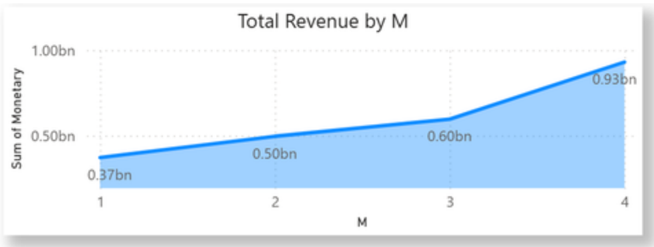
The chart shows that customer distribution across frequency groups is relatively balanced, ranging from approximately 25K to 30K customers.





The number of customers across monetary groups is almost evenly distributed, which reflects the quartile-based segmentation.





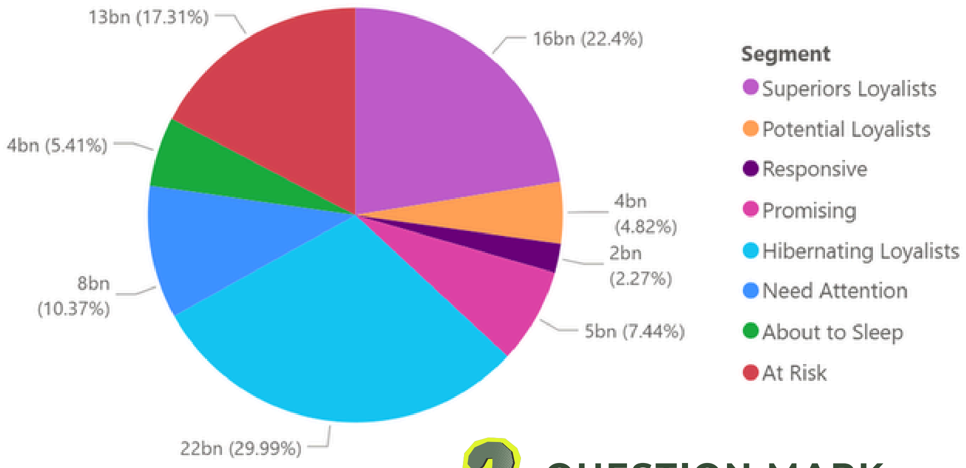
For Total Revenue charts for both F and M, although the number of customers in groups 3 and 4 is nearly the same, there is a significant gap in total revenue between these two groups. This indicates that average revenue per customer is significantly higher compared to group 3.
In other words, even a small improvement in customer behavior (moving from group 3 to group 4) can lead to a disproportionately large increase in revenue. This highlights that customer value is highly sensitive to spending and frequency at higher levels.

The business should focus on maximizing the value of high-performing segments. For group 4, strategies should prioritize retention through exclusive benefits, personalized experiences, and premium offerings to sustain their high spending behavior.

Beside that, **a small shift in customer behavior at higher segments can generate a disproportionately large impact on revenue**. For group 3, targeted incentives and upselling strategies can be implemented to encourage higher spending and move these customers into the top segment. By converting even a small portion of group 3 customers into group 4, the business can achieve significant revenue growth.

03 Evaluation based on customer segments

Speciafically, we will now explore the meaning of the four groups according to the BCG Matrix - the Growth-Share Metrix.



STAR

The Star group in the BCG Matrix refers to Superiors Loyalists. These customers have high purchase frequency, recent engagement, and strong spending behavior.
Currently, this group accounts for approximately 22.4% of total customers (~16K), making it one of the most valuable segments.

DOG

The Dog group includes About to Sleep and At Risk customers. These segments represent customers with low engagement and a high likelihood of churn.
Currently:

- About to Sleep: ~5.4%
- At Risk: ~17.3%

This indicates a significant portion of customers are at risk, especially the At Risk group, which alone accounts for a large share.

QUESTION MARK

The Question Mark group includes Potential Loyalists, Responsive, Hibernating Loyalists, and Need Attention. These groups typically score high in one of the RFM. These customers show potential but are not yet fully engaged across all dimensions.
This group represents the largest share of customers (~47–48%), including:

- Potential Loyalists: ~4.8%
- Responsive: ~2.3%
- Hibernating Loyalists: ~30.0%
- Need Attention: ~10.3%

CASH COW

The Cash Cow group corresponds to Promising customers, who account for approximately 7.4% of the customer base (~5K customers).
These customers have recent activity but moderate purchase frequency and spending levels.

The business is facing both growth and retention challenges, with a large number of customers at risk of churn; however, Question Mark customers still have untapped potential.

CHAPTER IV

SOLUTIONS FOR BUSINESS

Based on the analysis, the loyal customers contribute significantly to revenue, a large proportion of customers remain under-engaged or at risk of churn. Therefore, targeted strategies are required to both maximize customer value and improve retention.

FOR LOYAL CUSTOMERS (CASH COWS)

This group has stable engagement but has not yet reached peak value. From the analysis, increasing their activity can significantly boost revenue, such as, launch targeted promotions (bundles, discounts, cross-selling), encourage repeat purchases through loyalty programs, and recommend relevant products to increase basket size.

FOR SECONDARY CUSTOMERS (DOGS)

This group includes customers with low engagement and high churn risk. Business needs to focus on re-engagement and churn prevention, such as, run win-back campaigns, reconnect through personalized communication,..etc.

FOR VIP CUSTOMERS (STARS)

These customers generate the highest revenue and demonstrate strong engagement across Recency, Frequency, and Monetary metrics. To enhance retention and value maximization, business can provide exclusive benefits (VIP programs, early access, premium services), personalization to enhance customer experience.

FOR POTENTIAL CUSTOMERS (QUESTION MARKS)

This group represents a large portion of the customer base but lacks consistency across R, F, and M metrics. Among all segments, the Question Mark group presents the greatest growth opportunity due to its large size and untapped potential. This segment can be easily converted into high-value customers (Stars or Cash Cows). Therefore, targeted strategies for this segment should be a key focus to drive future revenue growth.

Business could apply personalized marketing campaigns, offer incentives to increase engagement and improve product relevance and customer experience

SQL CODE REFERENCE

```
WITH customer_statistic AS (
    SELECT
        ct.CustomerID
        , DATEDIFF('2022-09-01', MAX(STR_TO_DATE(ct.Purchase_Date, '%m/%d/%Y')))
          AS Recency
        , ROUND(COUNT(DISTINCT STR_TO_DATE(ct.Purchase_Date, '%m/%d/%Y'))
            / NULLIF(DATEDIFF('2022-09-01', STR_TO_DATE(cr.created_date, '%m/%d/%Y'))
            / 365.0,0),2) AS Frequency
        , ROUND(SUM(ct.GMV)/ NULLIF(DATEDIFF('2022-09-01',STR_TO_DATE(cr.created_date, '%m/%d/%Y'))
           / 365.0,0),2) AS Monetary
        , DATEDIFF('2022-09-01',STR_TO_DATE(cr.created_date, '%m/%d/%Y'))
          AS contract_age
    FROM Customer_Transaction ct
    JOIN Customer_Registered cr
        ON cr.ID = ct.CustomerID
    WHERE ct.Purchase_Date IS NOT NULL
      AND cr.created_date IS NOT NULL
    GROUP BY
        ct.CustomerID,
        cr.created_date
),

customer_rn AS (
    SELECT
        cs.*,
        ROW_NUMBER() OVER (ORDER BY cs.Recency ASC)   AS Recency_rn,
        ROW_NUMBER() OVER (ORDER BY cs.Frequency ASC) AS Frequency_rn,
        ROW_NUMBER() OVER (ORDER BY cs.Monetary ASC)  AS Monetary_rn,
        COUNT(*) OVER () AS total_rows
    FROM customer_statistic cs
),

cutoff AS (
    SELECT
        MAX(CASE WHEN Recency_rn   = ROUND(total_rows * 0.25, 0) THEN Recency   END) AS R_q1,
        MAX(CASE WHEN Recency_rn   = ROUND(total_rows * 0.50, 0) THEN Recency   END) AS R_q2,
        MAX(CASE WHEN Recency_rn   = ROUND(total_rows * 0.75, 0) THEN Recency   END) AS R_q3,

        MAX(CASE WHEN Frequency_rn = ROUND(total_rows * 0.25, 0) THEN Frequency END) AS F_q1,
        MAX(CASE WHEN Frequency_rn = ROUND(total_rows * 0.50, 0) THEN Frequency END) AS F_q2,
        MAX(CASE WHEN Frequency_rn = ROUND(total_rows * 0.75, 0) THEN Frequency END) AS F_q3,

        MAX(CASE WHEN Monetary_rn  = ROUND(total_rows * 0.25, 0) THEN Monetary  END) AS M_q1,
        MAX(CASE WHEN Monetary_rn  = ROUND(total_rows * 0.50, 0) THEN Monetary  END) AS M_q2,
        MAX(CASE WHEN Monetary_rn  = ROUND(total_rows * 0.75, 0) THEN Monetary  END) AS M_q3
    FROM customer_rn
),
```

SQL CODE REFERENCE

```
customer_rfm AS (
    SELECT
        cr.CustomerID,
        cr.Recency,
        cr.Frequency,
        cr.Monetary,

        CASE
            WHEN cr.Recency < c.R_q1 THEN 4
            WHEN cr.Recency < c.R_q2 THEN 3
            WHEN cr.Recency < c.R_q3 THEN 2
            ELSE 1
        END AS R,

        CASE
            WHEN cr.Frequency < c.F_q1 THEN 1
            WHEN cr.Frequency < c.F_q2 THEN 2
            WHEN cr.Frequency < c.F_q3 THEN 3
            ELSE 4
        END AS F,

        CASE
            WHEN cr.Monetary < c.M_q1 THEN 1
            WHEN cr.Monetary < c.M_q2 THEN 2
            WHEN cr.Monetary < c.M_q3 THEN 3
            ELSE 4
        END AS M

    FROM customer_rn cr
    CROSS JOIN cutoff c
)

SELECT
        *,
        CONCAT(R, F, M) AS RFM,
        CASE
            WHEN R >= 3 AND F >= 3 AND M >= 3 THEN 'Superiors Loyalists'
            WHEN R >= 3 AND F >= 3 AND M < 3 THEN 'Potential Loyalists'
            WHEN R >= 3 AND F < 3 AND M >= 3 THEN 'Responsive'
            WHEN R >= 3 AND F < 3 AND M < 3 THEN 'Promising'
            WHEN R < 3 AND F >= 3 AND M >= 3 THEN 'Hibernating Loyalists'
            WHEN R < 3 AND F >= 3 AND M < 3 THEN 'Need Attention'
            WHEN R < 3 AND F < 3 AND M >= 3 THEN 'About to Sleep'
            ELSE 'At Risk'
        END AS Segment
    FROM customer_rfm
```

DIEM PHUC LE

Data Analyst

Contact

+32-456134416

lediemphuc.nh@gmail.com

linkedin.com/in/phuc-le-thi-diem-05b45b143